Environmental Science and Pollution Research (2023) 30:47956–47971
https://doi.org/10.1007/s11356-023-25599-0

RESEARCH ARTICLE

Environment-oriented disassembly planning for end-of-life vehicle batteries based on an improved northern goshawk optimisation algorithm

Changshu Zhan[1] · Xuesong Zhang[1] · Guangdong Tian[2] · Duc Truong Pham[3] · Mikhail Ivanov[4] · Anatoly Aleksandrov[4] · Chenxi Fu[5] · Junnan Zhang[6] · Zhen Wu[6]

Received: 25 October 2022 / Accepted: 24 January 2023 / Published online: 7 February 2023

Abstract

Due to environmental pollution and resource shortages, the electric vehicle industry has been developing swiftly, and the market demand for batteries, as an essential part of electric vehicles, has also surged. Proper disassembly of end-of-life vehicle batteries (ELV batteries) is necessary to achieve the integrity and closure of their life cycle, promote the development of green remanufacturing, effectively reduce the pollution of the environment caused by metal ion leakage, and reduce people's dependence on natural resources to a certain extent. To schedule the disassembly operations of ELV batteries more rationally and further promote their disassembly quality and efficiency, this paper proposes a dual-objective disassembly sequence planning (DSP) optimisation model, which aims to minimise the hazard index and energy cost during ELV battery disassembly operations. Since the proposed model is a complex NP-hard optimisation problem, this study develops an efficient metaheuristic algorithm for solving this model based on the northern goshawk optimisation algorithm. The main algorithm adds two types of discrete recombination operators and a local search operator. At the same time, the predatory behaviour of the goshawk is optimised by combining the characteristics of the disassembly sequence planning problem to improve its performance. Finally, the disassembly of the battery of a Tesla Model 1 is used as a case study to demonstrate the effectiveness and feasibility of the proposed method.

Keywords Remanufacturing · Green manufacturing · Recycling of ELV batteries · Energy consumption · Disassembly sequence planning

Responsible Editor: Philippe Garrigues

✉ Guangdong Tian
tiangd2013@163.com

Changshu Zhan
zhchsh3@nefu.edu.cn

Xuesong Zhang
zz980818@126.com

Duc Truong Pham
d.t.pham@bham.ac.uk

Mikhail Ivanov
mivanov@bmstu.ru; president@bmstu.ru

Chenxi Fu
fuchenxi1010@163.com

Junnan Zhang
zjn087@163.com

Zhen Wu
13964648788@163.com

[1] Transportation College, Northeast Forestry University, Harbin 150040, China

[2] School of Mechanical-Electrical and Vehicle Engineering, Beijing University of Civil Engineering and Architecture, Beijing 100044, China

[3] Department of Mechanical Engineering, University of Birmingham, B15 2TT, Birmingham, UK

[4] Department of Ecological and Industrial Safety, Bauman Moscow State Technical University, Moscow, Russian Federation 105005

[5] School of Foreign Languages, Dalian Maritime University, Dalian 116000, China

[6] Department of Technical Development, Shandong Wina Green Power Technology Co., Ltd., Weifang 262700, China

Introduction

In recent years, the new electric vehicle industry has gradually gained attention, and the size of the market for electric vehicles is expanding daily (Tian et al. 2022a). With increasing national policy support, the new electric vehicle industry has been developing rapidly. In this context, ELV batteries are self-evident as a crucial component of electric vehicles. The technology of electric vehicle batteries is also becoming mature. However, the service life of a battery is usually only 5–6 years. After the capacity of batteries gradually decays, they will not be able to meet the demand of electric vehicle use, i.e., many batteries will face scrapping and recycling. Disassembling and recycling them can reduce environmental pollution, achieve resource and energy savings, and bring new economic growth points for regional economic development (Ma et al. 2021; Tian et al. 2022b). The rational scheduling of ELV battery disassembly is what motivated this paper.

Considering the structure of the ELV battery component and the problem characteristics of disassembly, the DSP problem is particularly suitable for disassembling ELV batteries. DSP is a vital issue in the product disassembly process. Its core idea is to generate optimal or near-optimal disassembly sequences with high efficiency under the condition of satisfying various disassembly constraints to achieve various disassembly goals (Tian et al., 2012), which can effectively improve the quality and efficiency of disassembly through DSP and promote the development of green remanufacturing, and this issue has received increasing attention since it was raised (Wang et al. 2021). The current state of development of DSP issues and the application of DSP in ELV batteries should be reviewed and analysed to develop a more effective approach to ELV battery DSP. The development of DSP problems involves the construction of disassembly models and their variants, the application of heuristic algorithms, and the current increasing the use of meta-heuristics. Suga et al. (1996) proposed a method for the quantitative assessment of disassemblability. Subramani and Dewhurst (1991) established the relationship model of the parts by analysing the assembly model of the product so that the disassembly sequence of the product could be generated automatically. Henrioud et al. (1991) represented the relationship between product structures in an associative graphical model. Sanderson and Homem de Mello (1990) et al. proposed a graphical model of AND/OR to represent the hierarchical relationship of parts to be disassembled in terms of two logical relationships, namely, AND or OR. The DSP is a typical discrete combinatorial optimisation problem. With the rapid progress of stochastic computer technology, many DSP studies tend to be solved by heuristic/metaheuristic algorithms (Yuan et al. 2020). Such methods sacrifice the quality of understanding to some extent but can save considerable computational time, thus compensating for the shortage of exact algorithms for solving large-scale complex DSPs (Bentaha et al. 2020). In the application of the heuristic algorithm, Gungor and Gupta (2001) introduced a branch delimitation heuristic that generates a reasonable and efficient disassembly solution, which uses a disassembly tree to describe the disassembly process of a product by establishing a priority matrix based on the priority constraints among the components in the product and then generating a feasible disassembly sequence with the help of the priority matrix. Moore et al. (2001) constructed a hybrid model combining the Petri net and a DSP model and designed a heuristic procedure to solve this model. Lambert and Gupta (2008) first found a better solution to the DSP quickly by the heuristic algorithm and then used the strategy of the iterative exact algorithm to further improve the quality of the solution. Due to their robustness, fast solution speed, and easy parameter adjustment (Tian et al. 2017), metaheuristic algorithms have become the main method for solving DSPs. Wang et al. considered the uncertainty of the disassembly time, cost, and part quality condition in the DSP and combined the artificial bee colony algorithm (ABC) optimisation process with fuzzy simulation to solve the optimal disassembly sequence under uncertain conditions (Wang et al. 2020). Zhu et al. (2020) established an improved frog leaping algorithm and proposed three-stage encoding to construct frog individuals to achieve diversity adjustment of selective disassembly sequences. Using a hybrid artificial fish swarming algorithm (AFSA) based on chaotic mapping and a genetic algorithm (GA), a novel hybrid method to find the optimal or near-optimal disassembly sequence was established by Guo et al. (2021). Chen et al. (2011) improved the chromosome encoding form and genetic operator of the basic genetic algorithm based on the characteristics of disassembly combination optimisation to obtain the optimal disassembly sequence. Xie et al. (2021) proposed an improved grey wolf optimisation algorithm (GWO) for solving the DSP and designed three new sequence optimisation approaches to ensure the solution under the complex constraint of disassembly priority feasibility. Tian et al. (2022b) used a social engineering optimisation algorithm (SEO) to solve the energy-efficient disassembly line balancing problem based on bucket brigades and cloud theory. Yuan et al. (2022) developed a metaheuristic based on the fruit fly optimisation algorithm (FOA) to find a fuzzy disassembly scheduling scheme. Zeng et al. (2022) studied the problem of energy-efficient and highly profitable robotic disassembly of used appliances with the aim of reducing the disassembly completion time for energy-efficient and highly profitable disassembly operations. Guo et al. (2021) developed a stochastic mixed discrete GWO for multiproduct DSP. Significantly, scholars have partially investigated the application of DSP in ELV battery disassembly. Xiao et al. (2022) proposed a dynamic Bayesian network-based disassembly sequencing method based on a disassembly graph model for ELV batteries, which provides manufacturers with dynamic process optimisation. In their study, Tan et al. (2021) proposed a hybrid disassembly framework for battery disassembly with a focus on improving disassembly efficiency. Alfaro et al. (2020) proposed a model for the disassembly process of battery packs for remanufacturing that provides the optimal level of disassembly and the most appropriate decision for the use of disassembled components. To establish a complete and open product information model and improve the efficiency of disassembly planning for ELV batteries, Yu et al. (2022) proposed a disassembly task planning method for automotive batteries based on ontology and partial destructive rules.

Through the above literature review, it is found that energy-saving DSP is uncommon. Most studies have focused on improving the efficiency and economy of disassembly operations, and there is also a lack of literature analysis for ELV battery DSP. Meanwhile, although there have been many metaheuristic algorithms applied to the DSP, such as ABC, AFSA, SEO, GWO, etc., the northern goshawk optimisation algorithm (NGO) has not been applied in the field of the DSP, which is simple, convenient, and easy to understand and has a powerful optimisation search capability (Dehghani et al. 2021). According to the no-free-lunch theorem (Wolpert et al. 1997), no optimisation algorithm can achieve the optimal solution for all problems, and it is necessary to apply it to DSPs (Feng et al. 2018).

To do so, this paper constructs a dual-objective ELV battery DSP model and improves the NGO by combining it with the DSP features. Compared with extant studies, this work has two contributions: (1) to better organise ELV battery disassembly operations, maximise their environmental benefits, and reduce the environmental hazard during their disassembly, a dual-objective ELV battery DSP model was developed to minimise both energy consumption and hazard index. (2) To solve the proposed DSP model, an improved NGO is reported with discrete recombination operators and a local search operator. The applicability of the proposed model and algorithm is demonstrated through the disassembly of an extensive instance and random cases of 8–80 parts. Figure 1 illustrates the research block diagram of this paper.

The rest of this paper is summarised as follows: Section 2 presents the developed dual-objective DSP optimisation model. Section 3 presents the improved multi-objective northern goshawk optimisation algorithm. Sections 4 and 5 provide example analysis and enhanced algorithm performance analysis. Finally, Section 6 summarises this paper's results, limitations, and recommendations.

Proposed problem

Here, the dual-objective DSP optimisation model is defined. The proposed issue is explained first in this regard to describe the idea of our DSP (Section 2.1). Finally, we deploy our dual-objective formulation of the DSP model (Section 2.2).

Problem description

In this research, a hybrid disassembly graph is introduced to describe the information related to disassembly. The hybrid disassembly graph consists of several nodes, undirected edges, and directed edges to describe the constraint relations of the product parts to be disassembled (Zhang et al. 2021). The disassembly hybrid graph can be presented in the triplet of Equation (1).

$$G = \{V, E, DE\} \tag{1}$$

where G is the hybrid diagram; V denotes parts, assuming there are n parts to be disassembled, then $V = \{V_1, V_2, V_3, V_n\}$; E is the undirected edge, indicating that there is a connection relationship between two parts; and DE is the directed edge, indicating that there is a compulsory priority constraint relationship between two parts of the product. A basic disassembly hybrid graph is shown in Fig. 2. The number in the circle indicates the task number, and the solid straight line indicates the undirected edge E. For instance, task 1 and task 5 are connected by a solid straight line, indicating that there is a connection relationship between these two tasks before. Solid arrows indicate that the directed edges DE, such as task 1 and task 4, are connected by solid arrows, which indicates that there is a compulsory priority constraint relationship before these two tasks. Task 1 must be completed before task 4.

To simplify the operation, the disassembly hybrid graph is transformed into a disassembly priority constraint relationship matrix P and a disassembly contact relationship matrix C (Zhang et al. 2021).

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} \quad c = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}$$

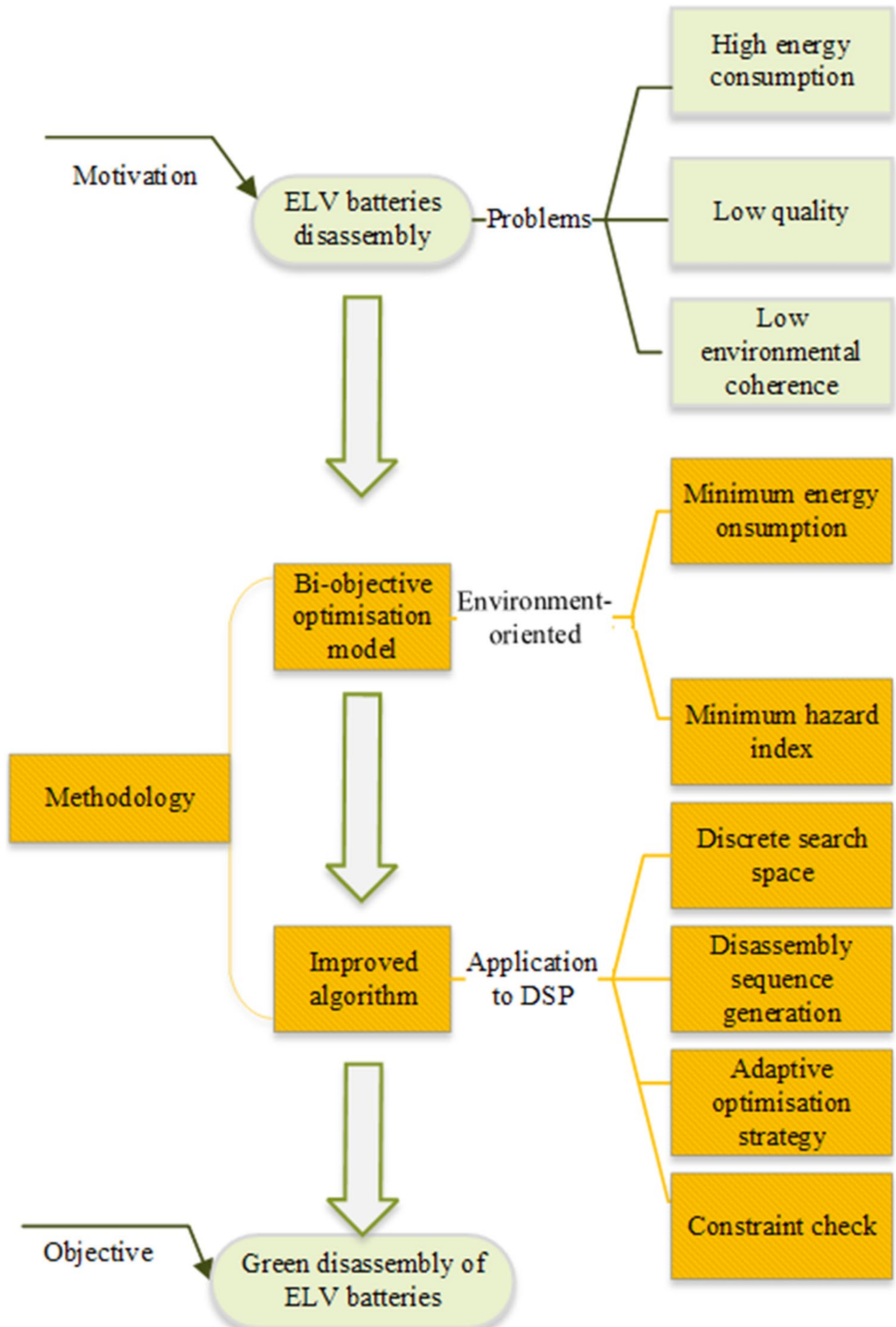


Fig. 1 General research block diagram

P_{ij}=1 when part i is the previous task of part j; otherwise, P_{ij}=0. As in Fig. 2, P_{14}=1.

When there is direct contact between part i and part j, C_{ij}=1. When they are not in direct contact or i=j, C_{ij}=0. As in Fig. 2, C_{13}=1.

When a part needs to be disassembled, it must be assessed for disassemblability, and the test formula can be expressed as Equation (2).

$$\sum_{j=1}^{n} c_{ij} = 1 \text{ and } \sum_{j=1}^{n} p_{ij} = 0 \tag{2}$$

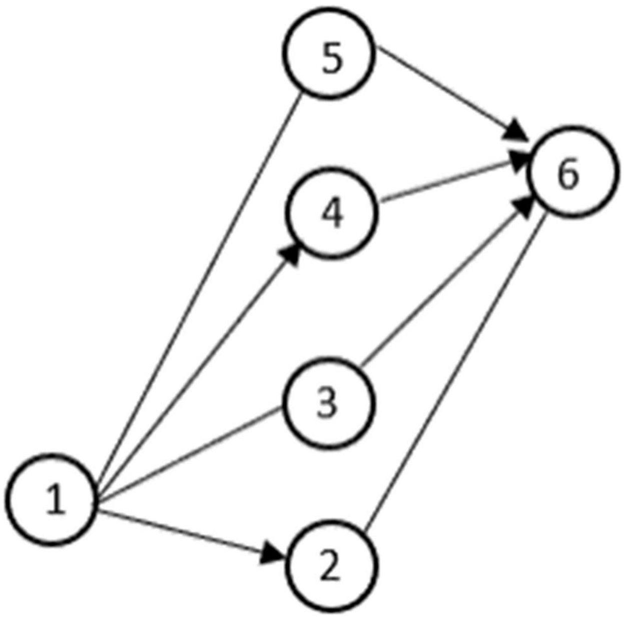


Fig. 2 Disassembled hybrid graph

where i represents the parts to be disassembled and n is the total number of disassembly tasks.

Proposed model

To minimise the hazards of disassembly operations for human beings and the environment and considering the importance of energy efficiency in the process of disassembly operations for ELV batteries, an optimisation model for ELV battery disassembly sequences was constructed in this study to minimise the hazard index and energy cost of disassembly operations. Combining the disassembly concept and the specific characteristics of ELV batteries, the disassembly tool and the number of disassembly direction conversions are represented in a unified dimension, and the following dual-objective ELV battery DSP optimisation model is constructed. To establish the proposed model, the following notations are used:

Indices:	
i/j:	Index of disassembly task number, $i/j = 1, 2, \ldots, n$
Parameters:	
n:	Total number of disassembly tasks
p_i:	Position of task i in the disassembly sequence
e_t:	Energy consumption generated by the change in the disassembly tool
w_t:	Weight of e_t
e_d:	Energy consumption generated by the change in the disassembly direction
w_d:	Weight of e_d
e_i:	Energy consumption generated by the disassembled part i
w_i:	Weight of e_i
g_i:	Disassembly task difficulty of task i
L	Fixed energy generated during the disassembly operation
Decision variables:	
h_i:	If part i is hazardous, h_i =1; otherwise, h_i =0.
t_{ij}:	If the disassembly tool in task i is different from task j, $t_{ij} = 1$; otherwise, $t_{ij} = 0$
d_{ij}:	If the disassembly direction in task i is different from task j, $d_{ij} = 1$; otherwise, $d_{ij} = 0$

$$\min F_{DSP} = [f_1, f_2] \tag{3}$$

$$f_1 = \sum_{i=1}^{n} (p_i \cdot h_i) \tag{4}$$

$$f_2 = \sum w_t t_{ij} e_t + \sum w_d d_{ij} e_d + \sum w_i (1 + g_i) e_i + L \tag{5}$$

$$t_{ij}, d_{ij}, h_i = \{0, 1\} \tag{6}$$

Equation (3) represents the hazard index and energy consumption during the disassembly operation of the minimised battery system. Equation (4) represents the formula for calculating the hazard index generated by the disassembly operation, p_i is the position of task i in the disassembly sequence, and h_i is a 0–1 variable indicating whether part i has a hazard attribute. If the part is hazardous, then h_i=1; otherwise, h_i=0. Equation (5) represents the integrated energy consumption generated during the disassembly process, e_t is the energy consumption generated by the change in the disassembly tool, with a weight of w_t; e_d is the energy consumption generated by the change in disassembly direction, weighted as w_d; e_i is the energy consumption generated by the disassembled part itself, weighted as w_i; g_i is the disassembly task difficulty of task i; L is the fixed energy consumption generated by the whole disassembly operation process; t_{ij} and d_{ij} denote 0–1 variables, t_{ij}=1 when the disassembly tool is changed and in addition t_{ij}=0 and d_{ij}=1 when the disassembly direction is changed and in addition d_{ij}=0; and L is the fixed energy generated during the disassembly operation.

Proposed solution method

Here, the improved NGO is presented as a solution to the proposed problem, and we first introduce the traditional NGO algorithm (Section 3.1). Later, we outline the core idea of the proposed improved NGO based on the problem characteristics and concept of the traditional NGO and define its main framework to demonstrate how these elements are connected in our improved NGO (Section 3.2).

Traditional NGO

The NGO proposed by Dehghani et al. in 2021 simulates the behaviour of the northern goshawk during prey hunting. NGO has the advantages of fast convergence and high stability (Dehghani et al. 2021). Nowadays, NGO has been applied to a variety of fields and is proven to have good performance in solving problems, such as photovoltaic model parameter identification, neural network optimization, and vehicle scheduling. Experimental results show that NGO has certain advantages over algorithms such as GA and PSO. Its hunting strategy consists of prey identification and the tailing process.

Prey identification and prey attack (global search)

In the first hunting stage, the northern goshawk randomly selects its prey and then quickly attacks it, belonging to the global search phase, which can be described by the following mathematical model (Dehghani et al. 2021).

$$P_i = X_k, i = 1, 2, \ldots, N, k = 1, 2, \ldots, i-1, i+1, \ldots, N \tag{7}$$

$$x_{i,j}^{\text{new},P1} = \begin{cases} x_{i,j} + r\left(p_{i,j} - Ix_{i,j}\right), & F_{P_i} < F_i, \\ x_{i,j} + r\left(x_{i,j} - p_{i,j}\right), & F_{P_i} \geq F_i, \end{cases} \tag{8}$$

$$X_i = \begin{cases} X_i^{new,P1}, & F_i^{\text{new},P1} < F_i, \\ X_i, & F_i^{\text{new},P1} \geq F_i, \end{cases} \tag{9}$$

where P_i is the position of the ith northern goshawk's prey, F_{pi} is the objective function value of the position of the ith northern goshawk's prey, k is a random integer in the range [1, N], $X_i^{new,P1}$ is the new position of the ith northern goshawk, $x_{i,j}^{\text{new},P1}$ is the new position of the jth dimension of the ith northern goshawk, $F_i^{\text{new},P1}$ is based on the objective function value of the ith northern goshawk after the phase 1 update, and r is a random number in the range of [0, 1].

Chase and escape (local search)

After a northern goshawk attacks its prey, the prey will attempt to escape. Therefore, the northern goshawk needs to continue to chase the prey during the closing of the chase. Due to the high pursuit speed of the northern goshawks, they can chase prey in almost any situation and eventually capture prey. The simulation of this behaviour improves the algorithm's ability to search the search space locally. This hunting activity is assumed to be close to an attack position with a radius of R. In the second stage (Dehghani et al. 2021), it is described by Equations (10) to (12).

$$x_{i,j}^{new,P2} = x_{i,j} + R(2r-1)x_{i,j} \tag{10}$$

$$R = 0.02\left(1 - \frac{M}{Maxit}\right) \tag{11}$$

$$X_i = \begin{cases} X_i^{\text{new},P2}, & F_i^{\text{new},P2} < F_i \\ X_i, & F_i^{\text{new},P2} \geq F_i \end{cases} \tag{12}$$

where m is the current number of iterations, $Maxit$ is the maximum number of iterations, $X_i^{\text{new},P2}$ is the new position of the ith solution, $x_{i,j}^{new,P2}$ is its jth dimensional value, and $F_i^{\text{new},P2}$ is the objective function value of the second stage of the NGO.

Improved NGO

NGO has been shown to have advantages and to be effective in handling different optimisation problems (Dehghani et al. 2021). However, it needs to be modified to solve complex optimisation problems such as the proposed DSP. The first issue is that the proposed DSP is a multi-objective optimisation problem, but NGO applies to single-objective problems. Therefore, NGO needs to be extended to allow it to solve multi-objective optimisation problems. The second issue is that the search space of the NGO is continuous and NGO cannot solve integer optimisation problems. However, the feasible space of the DSP problem that needs to be solved in this paper is the constraint-compliant disassembly sequence, and therefore, a discrete search space and optimisation rules for improving NGO are required. Combining the traditional NGO concept and the characteristics of the DSP, this research improves NGO so that it can solve discrete problems. The main ideas of the improved NGO are as follows:

- Based on the traditional NGO's dual phase of global search and local search
- An encoding approach based on the DSP characteristics
- Disassembly sequence generation and checking and optimisation methods
- Introduction of Pareto dominance idea and crowding distance mechanism for multi-objective problems

The encoding method based on the DSP characteristics

The search space of the traditional NGO is continuous, while the DSP to be solved is a discrete space. Therefore, this

1	5	2	4	3	6

Fig. 3 Disassembly sequence representation

study introduces an encoding method based on the characteristics of the DSP to optimise the traditional NGO search space so that it can solve discrete optimisation problems. The method is performed in a real number encoding (1, 5, 2, 4, 3, 6), indicating the disassembly order: 1-5-2-4-3-6. Figure 3 shows a representation of the disassembly sequence.

Disassembly sequence generation and checking methods

The DSP belongs to the strong constraint problem, and the initial disassembly sequence needs to satisfy the constraint relationship between the parts. Based on the disassembly priority constraint relationship matrix and the disassembly contact relationship matrix, the initial disassembly sequence formation method is proposed in combination with this study's proposed coding method.

First, two matrices are generated based on the disassembly hybrid graph. Second, the matrix rows that meet the disassemblability are found, and one of the rows is randomly selected as the first part to be disassembled. Then, the row is removed from the matrix, and the above steps are repeated until the disassembly priority constraint relationship matrix and the disassembly contact relationship matrix are empty, thus generating a feasible disassembly sequence.

The optimised disassembly sequence requires checking whether it satisfies disassemblability, starting from the first position in the disassembly sequence. If this disassembly task can be completed, the disassemblability of the next task is checked. If it cannot be completed, a randomly selected completable task is used to replace it, and then the disassemblability of the next task is checked. The above process is repeated until the last task satisfies the disassembly requirements.

Pareto dominance and crowding distance mechanisms

The multi-objective problem is treated by introducing Pareto dominance and the crowding distance mechanism. Pareto dominance is used to treat the minimisation problem with K sub-objectives, and for any two solutions, L_1 and L_2, in the solution space, L_1 dominates L_2 if the relationship in Equation (13) is satisfied (Gholizadeh et al. 2021; Fathollahi-Fard et al. 2021).

$$\begin{cases} \forall k \in \{1,2,\cdots,K\},\ f_i(L_1) \leqslant f_i(L_2) \\ \exists k \in \{1,2,\cdots,K\},\ f_j(L_1) < f_j(L_2) \end{cases} \tag{13}$$

The crowding distance mechanism filters the disassembly sequences that exceed the number of external files and eliminates the crowded region solutions to obtain the approximate Pareto optimal front with uniform distribution. Set the bounded solution $Cd_1 = Cd_q = \infty$, and calculate the crowding distance for the solutions with $i = 1,2,3$, q-1, which is calculated as follows:

$$Cd_i = \begin{cases} \infty, & i = 1, q \\ \sum_{k=1}^{l} \frac{f_k^{i+1} - f_k^{i-1}}{f_k^{\max} - f_k^{\min}}, & i = 2, 2, \cdots, q-1 \end{cases} \tag{14}$$

where q is the number of non-inferior solutions. $f_k^{\max}$ and $f_k^{\min}$ are the minimum and maximum values of the Kth objective, respectively. Cd_i is the crowding distance of the objective function f_i, and l is the number of sub-objectives.

Finally, the improved NGO steps are described as follows:

Step 1: initialise the disassembly sequence The initial disassembly sequence satisfying the detachability is formed according to the proposed method to facilitate the optimisation of subsequent iterations.

Step 2: initial disassembly sequence discrete recombination To improve the quality of the initial disassembly sequence, this study introduces two types of discrete recombination operators, one of which is randomly selected to optimise the initial disassembly sequence after it is formed, after which Pareto domination and crowding distance are calculated for the optimised disassembly sequence and the initial disassembly sequence to update the initial disassembly sequence and select the optimal disassembly sequence. The restructuring methods are as follows:

Fig. 4. First discrete recombination method.

(1) Randomly select two points in the initial disassembly sequence, and cross them, as illustrated in Fig. 4.

(2) Randomly select two points in the initial disassembly sequence and uniquely place them in the last two of the initial disassembly sequences, as depicted in Fig. 5.

Step 3: prey identification and prey attack (global search) In this stage, this study also proposes two types of disassembly sequence optimisation, by which the disassembly sequence to be optimised can be fused with the optimal disassembly sequence selected in the second step while effectively improving the disassembly sequence diversity.

(1) A segment is randomly selected in the disassembly sequence to be optimised, and the disassembly order of the segment task is found in the optimal disassembly sequence selected in the previous step. It is updated according to that order, as depicted in Fig. 6.

(2) A segment of the optimal disassembly sequence selected in the second step is randomly selected, and the same position of this segment is found in the disassembly sequence that needs to be optimised and replaced, as shown in Fig. 7.

The two global search strategies are selected according to the adaptive probability R. The formula for calculating R is shown in Equation (11).

When $|R|>=0.01$, the first global search strategy is chosen to update the disassembly sequence; when $|R|<0.01$, choose the second global search strategy to update the disassembly sequence.

Fig. 5. Second discrete recombination method.

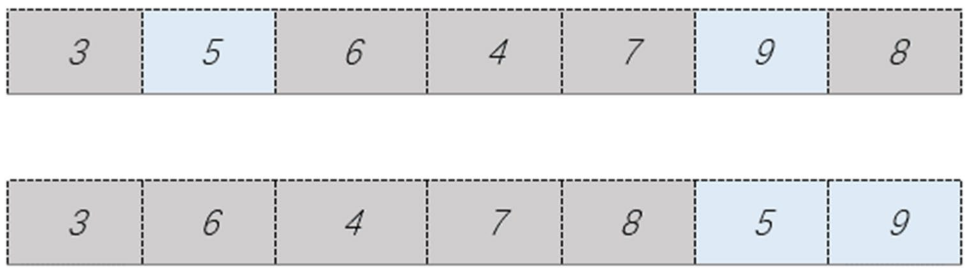


Fig. 6 First global search strategy

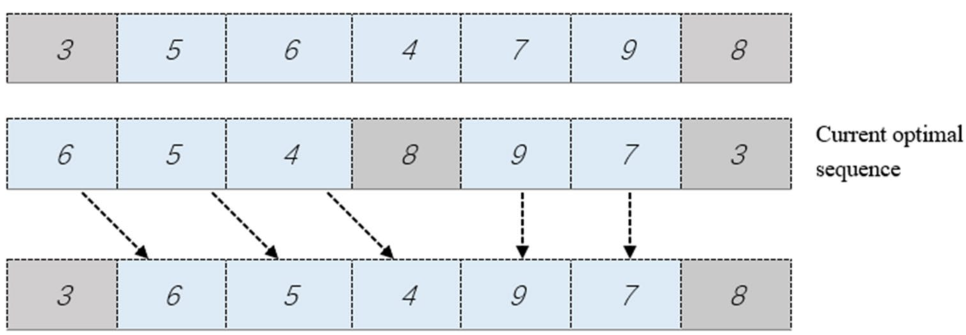
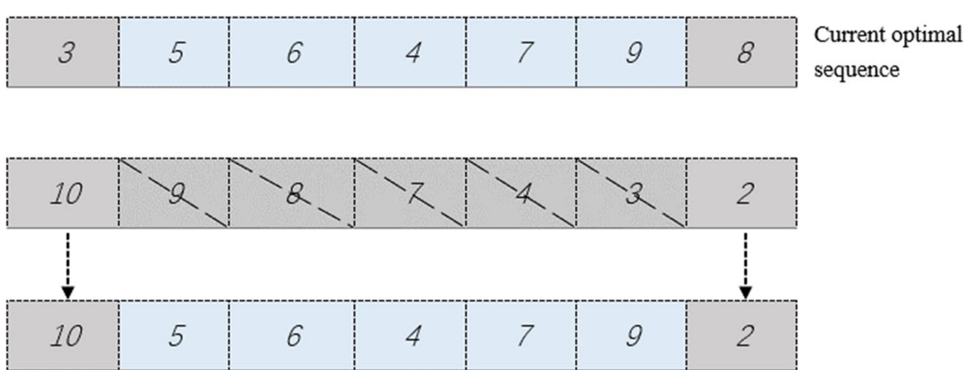


Fig. 7 Second global search strategy



The optimised disassembly sequence is evaluated for detachability, and the optimised disassembly sequence is replaced if its fitness is higher than that of the original disassembly sequence.

Step 4: chase and escape (local search) In this stage, a local search operator is introduced, and the method of generating random numbers is used to optimise the disassembled sequences, such as generating random numbers of six. Any six of them are selected for optimisation. The optimisation is performed as follows:

To optimise the disassembly sequence, the number corresponding to its disassembly task number is calculated to generate its corresponding disassembly sequence. The corresponding sequence generation method for a disassembly sequence of five tasks is given in Fig. 8.

The optimised disassembly sequence is tested for detachability, and the optimised disassembly sequence is replaced if its fitness is higher than that of the original disassembly sequence.

Step 5: Pareto dominance and crowding distance calculation After the optimisation is completed, the Pareto domination and crowding distance are calculated for the disassembly sequence, the Pareto first-level solution is saved to the external archive, and the optimal disassembly sequence is updated according to the crowding distance. The external file is updated for optimisation after each iteration. The optimal solution is found until the maximum number of iterations is reached.

Finally, the flow chart of the improved NGO proposed in this paper is shown in Fig. 9.

Results and discussion

Here, we extend the proposed model and algorithm to an example application. First, we provide an explanation of our case study to demonstrate its usefulness (Section 4.1). Subsequently, the proposed improved NGO parameters are calibrated to increase its effectiveness (Section 4.2). In the end, the case was resolved and discussed (Section 4.3). It should be noted that the experimental program of the proposed algorithm was developed using MATLAB R2020b on a computer with Intel(R) Core(TM) i5-7400 CPU at 3.00.

Case study

To verify the model's validity and algorithm, the battery of a Tesla Model 1 (Liu et al. 2021) is used as the research object, and the disassembly hybrid graph is generated according to the connection relationship and priority disassembly relationship between the battery parts, as shown in Fig. 10. The information on the battery disassembly system is shown in Table 1.

Parameter calibration

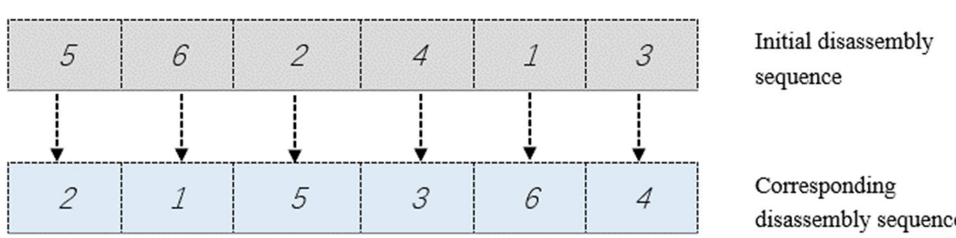
For more flexibility in decision-making, the number of external archives was set to 10 to increase the number of solutions. In addition, considering the practicalities of Tesla Model 1 battery disassembly and disassembly experience, combined with literature analysis (Tian et al. 2011a, 2011b), the following parameter values were used: w_t=1, w_d=0.8, w_i=1, e_t=5, e_d=3, and L=100. For the remaining two parameters, the initial population size (*Npop*) and the number of iterations (*Maxit*), to ensure the efficiency of the algorithm, four reference values were provided for each parameter, after which 16 experiments were performed. Table 2 displays the parameters and their reference values, and the reference values for each parameter were combined in pairs.

In this section, the difference *PE* between the mean of the sum of the initial objective function values and the mean of the sum of the final objective function values is used as an evaluation criterion for parameter selection, with a larger *PE* indicating better algorithm performance. The formula for *PE* is shown in Equation (15). Due to the random nature of the metaheuristic algorithm, each experiment was performed fifteen times, and the average was taken as the final result. The results of the 16 experiments are presented in Table 3.

$$PE = \frac{\sum(f_{a1}+f_{a2}+f_{a3})}{NS}initial - \frac{\sum(f_{a1}+f_{a2}+f_{a3})}{NS}final\ \alpha = 1,2,\ldots,NS \quad (15)$$

where f_{a1}, f_{a2}, and f_{a3} are the objective function values and *NS* is the number of solutions.

Fig. 8. Local search strategy.



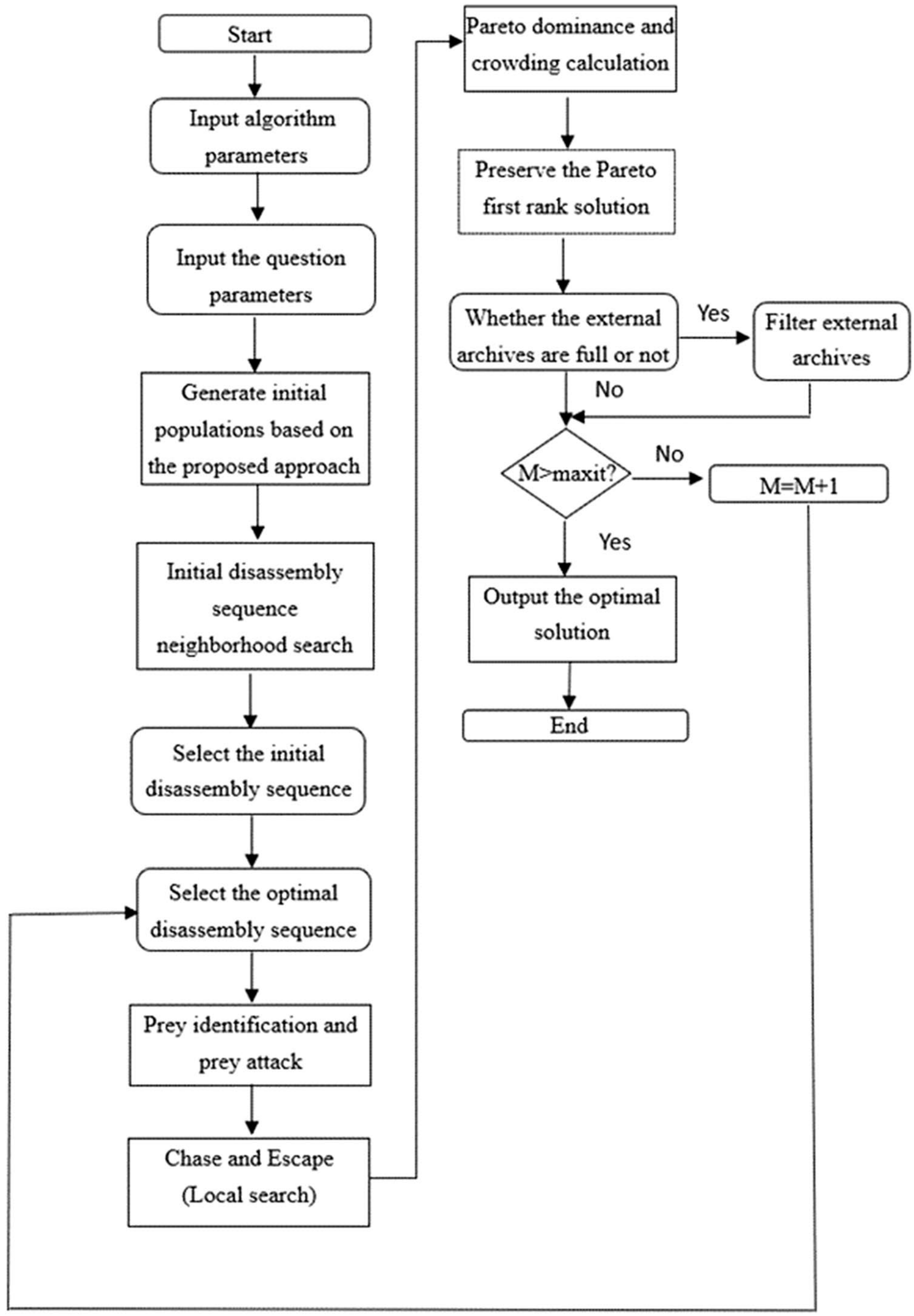


Fig. 9 Flow chart for the improved NGO

According to the results in Table 3, the *PE* was taken to the appropriate value in the 11th experiment, and although it still increases as the population size and the number of iterations increase, it is not significantly different, and given the efficiency of the algorithm, the final parameters were set to w_t=1, w_d=0.8, w_i=1, e_t=5, e_d=3, L=100, $Npop$=50, and $Maxit$ =200. The size of the external archive was 10.

Results and analysis

Using the information in Table 1, the disassembly sequence obtained for the Tesla Model 1 battery can be performed according to the methodological steps proposed in the previous section. Table 4 shows the set of external archives after the program has been run once.

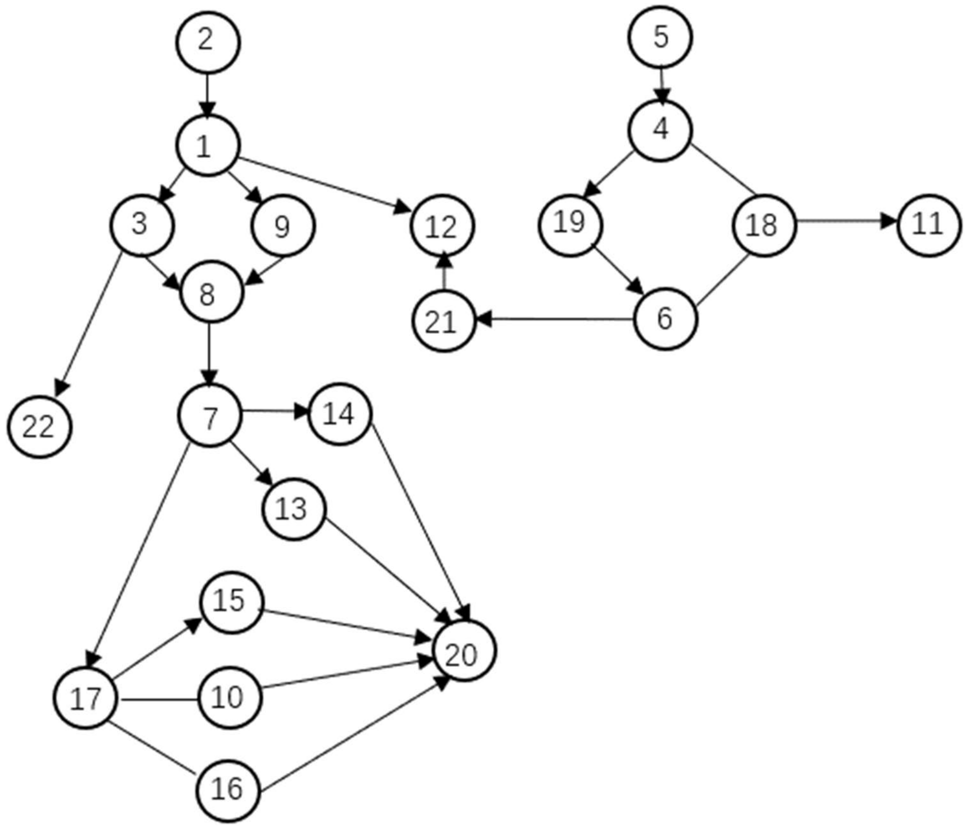


Fig. 10 The Tesla Model 1 battery disassembly hybrid graph

Table 1 Tesla Model 1 battery disassembly information

Order	Name	Unit time energy consumption	Tool	Disassembly difficulty	Disassembly time	Direction	Hazard index
1	Upper shell	0.0524	Spanner	0.25	180	+z	0
2	Upper shell screws	0.0203	Screwdriver	0	51	+z	0
3	Acoustic foam	0.0304	Plier	0.25	45	+z	1
4	High-pressure assembly upper shell	0.0418	Plier	0	30	+z	1
5	High-pressure assembly on shell screw	0.0291	Screwdriver	0	45	+z	0
6	Fuses	0.0244	Hand	0.25	24	+z	1
7	Battery modules	0.0356	Plier	0	60	+z	1
8	Busbars	0.0402	Plier	0	45	+y	1
9	Metal spacers	0.0365	Plier	0.5	42	+y	0
10	Insulation mats	0.1408	Hand	1	18	+z	0
11	Fibre boards	0.0280	Plier	0	9	+z	0
12	Cooling lines	0.0481	Hand	0	33	-y	0
13	Charging port connectors	0.1685	Plier	1	21	-y	1
14	Battery harnesses	0.3746	Hand	0.5	24	-y	1
15	Battery Management Systems BMS	0.0426	Hand	0.75	12	-x	1
16	Battery system harnesses	0.0492	Plier	0.5	24	-x	1
17	Module PCBs	0.0362	Plier	0	24	-x	0
18	Fibreboard screws	0.0321	Screwdriver	0	15	-x	0
19	High-voltage electrical fuses	0.0274	Screwdriver	0	6	+x	0
20	Electric cores	0.0269	Hand	0	48	+x	1
21	Coolants	0.0362	Plier	0.025	36	+x	0
22	Harmful battery packs	0.1874	Plier	1	24	+x	1

Table 2 List of parameters of the improved NGO algorithm

Parameters	Level 1	Level 2	Level 3	Level 4
Npop	10	30	50	70
Maxit	50	100	200	250

Table 3 Calibration experiments for the improved NGO

Number of experiments	Parameters and their levels		*PE*
	Npop	*Maxit*	
L1	1	1	36.86
L2	1	2	38.63
L3	1	3	40.18
L4	1	4	42.62
L5	2	1	39.38
L6	2	2	43.52
L7	2	3	45.23
L8	2	4	47.26
L9	3	1	43.68
L10	3	2	45.68
L11	3	3	48.76
L12	3	4	48.82
L13	4	1	44.23
L14	4	2	46.56
L15	4	3	48.80
L16	4	4	48.85

As seen from Table 4, the number of tool and direction changes in the sequence planning process of ELV batteries has an impact on their operation, and the number of direction and tool changes should be reduced as much as possible to maximise the efficiency and quality of ELV battery disassembly operations. In addition, among the obtained disassembly sequence options, the energy consumption is in the range 233.54–277.94, and the hazard index is in the range 107–144. If choosing option 5 can minimise the energy consumption generated in the process of disassembling ELV batteries and choosing option 2 can minimise the hazard index, the most balanced option for the two objectives is option 6. This study provides decision makers with more scope to consider how to better harmonise ELV battery disassembly operations with nature. ELV batteries have serious safety hazards and cause harm to the surrounding environment and society if not properly disposed of. Disassembly is an important means of recycling. Minimising the energy consumption and hazards generated during the disassembly of ELV batteries can effectively improve the quality and efficiency of ELV battery disassembly, enhance its overall disassembly process to be more in harmony with the natural ecology, offer new ideas for disassembly operations, and advance the development of green remanufacturing.

Comparison with other algorithms

Here, the performance of the improved NGO is carefully examined, firstly by making an extensive performance comparison with a number of other state-of-the-art algorithms to demonstrate the effectiveness of the proposed improved NGO (Section 5.1), followed by case studies with 8–80 parts to demonstrate its effectiveness for disassembly operations of different sizes (Section 5.2).

Comparison of case results

To evaluate the algorithm's performance, a comparison was conducted between the improved NGO, non-dominated sorting genetic algorithm II (NGSA-II), GWO, and artificial fish swarm algorithm (AFSA). The population sizes were 20, 30, 40, and 50, and the number of iterations was 200. The parameter settings for the three algorithms were based on the literature (Tian et al. 2022a, 2022b; Xie et al. 2021; Wang

Table 4 The elite set of the instance

Order	Disassembly sequences	Number of tool changes	Number of direction changes	f_1	f_2
1	16,2,5,1,3,4,9,8,7,13,17,22,19,6,15,18,21,12,14,10,20,11	9	13	243.54	120
2	16,2,1,9,3,8,7,13,22,14,17,10,15,20,5,4,19,6,18,11,21,12	13	19	277.94	107
3	5,2,1,16,3,9,8,22,7,13,4,18,19,10,6,14,11,17,21,12,15,20	6	17	238.14	128
4	16,2,1,9,3,8,5,22,7,14,13,4,17,19,615,10,20,21,12,18,11	13	18	275.54	111
5	16,5,4,19,18,2,16,10,11,3,22,9,21,8,7,13,17,15,14,12,20	7	13	233.54	144
6	16,2,5,1,4,3,22,9,8,7,13,17,15,14,19,18,11,6,21,10,12,20	9	14	245.94	116
7	2,1,3,9,16,8,22,7,17,15,14,10,13,20,5,4,18,19,6,11,21,12	11	16	260.74	112
8	16,2,1,5,3,9,8,22,7,13,4,18,19,10,6,14,11,21,17,15,12,20	8	15	243.34	124
9	2,1,5,3,16,9,4,22,19,18,6,11,8,7,21,17,13,14,12,15,10,20	7	14	235.94	139
10	16,2,5,1,3,9,22,8,7,13,17,15,14,10,4,20,19,18,6,11,21,12	10	14	250.94	115

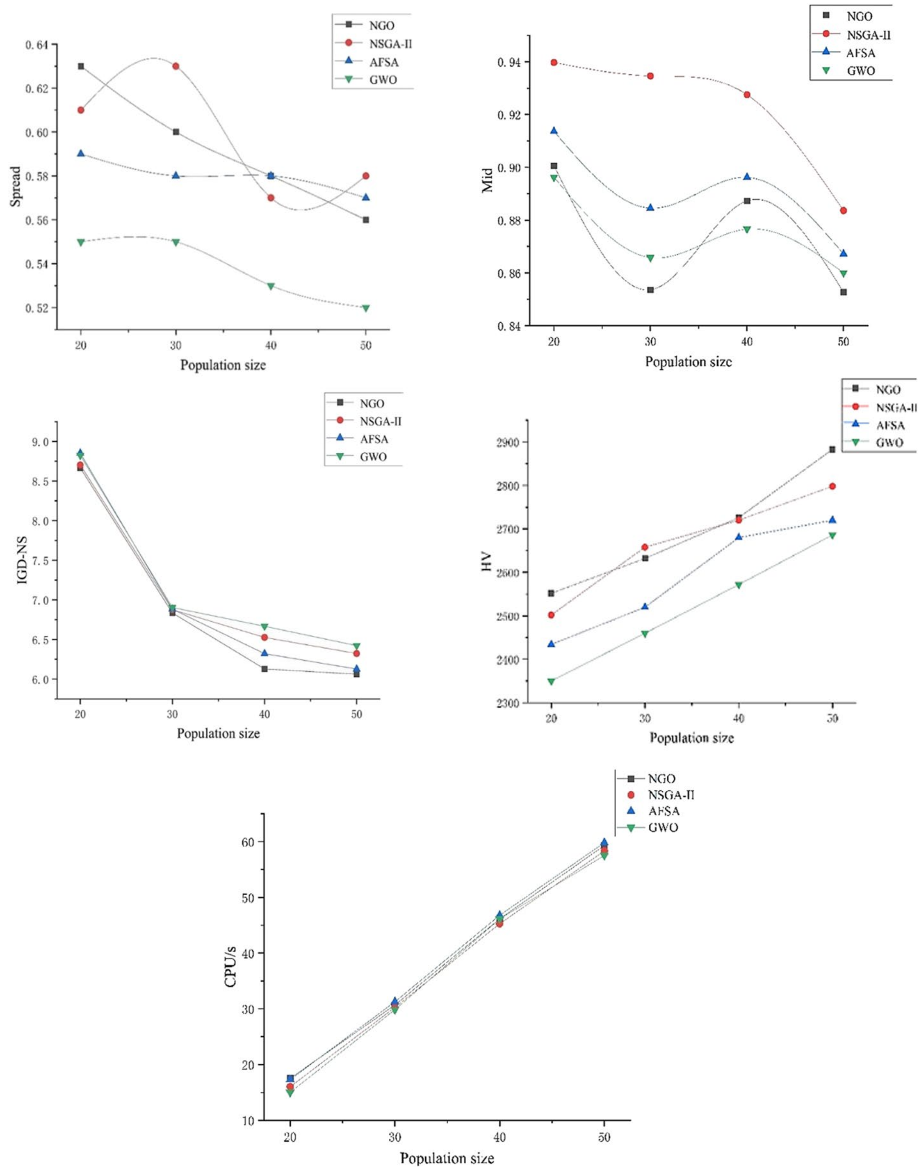


Fig. 11 Results of algorithm performance comparison

et al. 2020; Guo et al. 2021), and the problem parameters were the same as in the previous section. Since the problem was a complex optimisation problem and the actual Pareto front was unknown, the optimal populations obtained by the above algorithms were combined to obtain the approximate Pareto front, and the mean ideal distance (MID), enhanced inverse generation distance (IGD-NS), and spread indicator (Spread) were calculated accordingly. MID reflects the partial diversity of the algorithm, IGD-NS indicates the distance of each point of the true front from the known front nearest Euclidean distance, which reflects the convergence and diversity of the algorithm, and Spread reflects the distribution of the Pareto optimal solution obtained by the algorithm on the true Pareto front. In addition, the reference point was set to (1,1) to calculate the hypervolume (HV) to evaluate the overall performance of the algorithm. Except for HV, the smaller the metric, the better the algorithm performance. Since the metaheuristic algorithm was randomised, reliable results were obtained by running each algorithm fifteen times. The results are shown in Fig. 11. Figure 12 presents the statistics for a population size of 50.

According to the results in Fig. 11, it can be seen that although the improved NGO does not have a great advantage when the population size is small, as the population size increases, the improved NGO obtains optimal values for MID, HV, and IGD-NS, proving its excellent overall performance. In addition, it should be noted that the proposed initial population reorganisation operator increases the CPU time of the improved NGO, but the comparison shows that there is no great disadvantage in its CPU time, and its increase in running time can improve the quality of the solution set. Fig. 12 also demonstrates the good robustness and accuracy of the improved NGO.

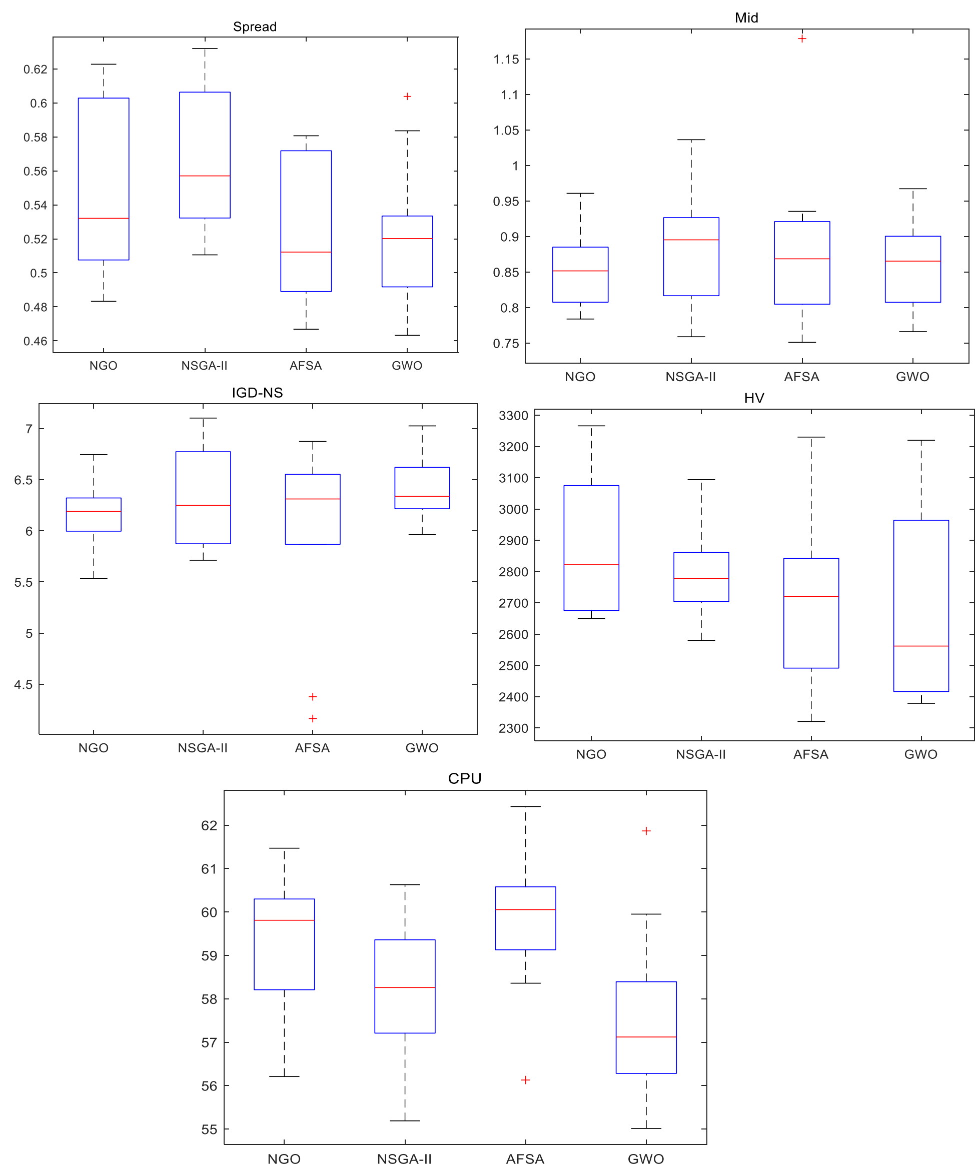


Fig. 12 Algorithm performance comparison box plots

Cases with 8–80 parts

To assess the practicality of the algorithm proposed in this study at various scales, random cases with 8–80 parts from McGovern et al. (2006) were solved using the improved NGO and compared with their solution results, with the solution objectives being the disassembly operation hazard index f_3 and the disassembly operation demand index f_4. The data were generated as shown in McGovern et al. (2006), with the theoretical optimal solution f_3=1, f_4=2 or f_3=2, f_4=1. When the sum of f_3 and f_4 is 3, the theoretical optimal solution of this multi-objective problem is obtained. The solution results are shown in Table 5.

Table 5 and Fig. 13 show that the improved NGO can produce excellent solutions to problems, resulting in the ideal solution for 12 of them. As the product structure became more complex, although it did not achieve the ideal solution to the problem, the results obtained were less different from the ideal solution, and its solution curve was infinitely close to the ideal solution curve. In addition, when dealing with disassembly operations of all sizes, the NGO shows better applicability compared to the ACO, with its solution curves all lying below the ACO, which does not have an optimal solution for any of the problems.

In summary, it can be concluded that the proposed improved NGO can solve the DSP effectively, has better solving efficiency and accuracy, can find the approximate Pareto front better, and saves computational resources.

Conclusion and future works

Disassembly is a critical part of recycling ELV batteries, and to increase the efficiency and quality of disassembly, reduce recycling costs, and promote environmental coherence, a proper

Table 5 Comparison of two algorithms for solving random cases

Number of parts	ACO		NGO	
	f_3	f_4	f_3	f_4
4	8	6	1	2
8	7	3	1	2
12	6	3	1	2
16	4	13	1	2
20	11	8	2	1
24	10	13	2	1
28	13	14	1	2
32	12	11	1	2
36	15	22	1	2
40	7	18	1	3
44	17	27	1	2
48	4	26	1	3
52	26	36	1	2
56	16	28	3	1
60	8	36	1	3
64	13	40	3	1
68	8	38	1	2
72	5	36	1	3
76	3	28	1	3
80	8	6	1	3

disassembly sequence is vital. This highlights the importance of the DSP. However, the current DSP modelling of ELV batteries is more concerned with cost and efficiency and cannot take into account different standards. The dual-objective ELV battery DSP model proposed in this paper simultaneously minimizes the energy use and hazard indices produced while ELV batteries are being disassembled. In addition, since the proposed problem is a complex optimisation problem, this study proposes a new extension of the traditional NGO to optimise the proposed model. Combining the discrete characteristics of DSP and the structural characteristics of ELV batteries, the coding and optimization rules of the algorithm are redesigned, and discrete recombination operators and a local search operator are introduced to improve the search performance of the algorithm. The validity of the model and algorithm is verified by examples of Tesla Model 1 and cases at different scales.

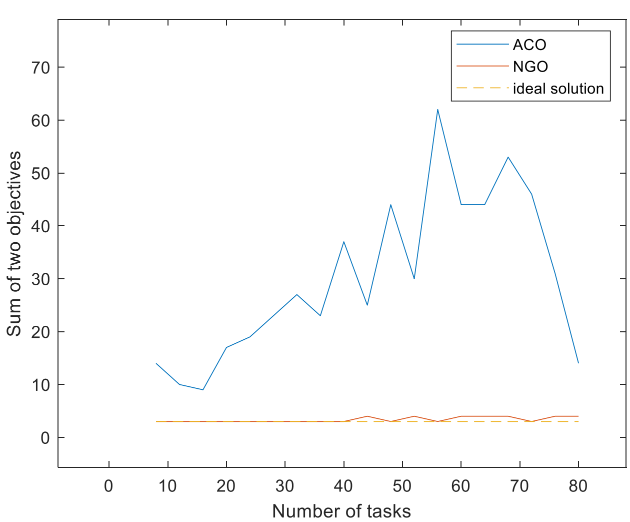


Fig. 13 Results of the random case solutions

Although this paper has studied the DSP of ELV batteries, there is still a large amount of unexplored research space. The introduction of robots (Laili et al. 2021; Wu et al. 2022) into the serial planning of ELV battery disassembly in conjunction with the integration of multicriteria decision-making techniques (Feng et al. 2019), the consideration of fuzzy uncertainty disassembly and the introduction of more constraint relationships in disassembly modelling to investigate more complex ELV battery disassembly operations are future research directions that can be further explored. Finally, the proposed improved NGO can be analysed using more performance metrics and other disassembly examples and extended by employing adaptive memory strategies and taboo search tools.

Author contributions Changshu Zhan: conceptualisation, methodology, funding acquisition, resources, supervision, validation, writing—original draft and editing. Xuesong Zhang: conceptualisation, methodology, formal analysis, writing—original draft. Guangdong Tian: methodology, formal analysis. Duc Truong Pham and Chenxi Fu: conceptualisation, supervision, methodology, validation. Mikhail Ivanov and Anatoly Aleksandrov: conceptualisation, supervision, methodology, validation. Junnan Zhang and Zhen Wu: supervision.

Data Availability Not applicable.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

Alfaro-Algaba M, Ramirez FJ (2020) Techno-economic and environmental disassembly planning of lithium-ion electric vehicle battery packs for remanufacturing. Resour Conserv Recycl 154:104461

Bentaha ML, Voisin A, Marangé P (2020) A decision tool for disassembly process planning under end-of-life product quality. Int J Prod Econ 219:386–401

Chen J, Zhang Y, Liao H (2011) Disassembly sequence planning based on improved genetic algorithm. In: Advances in Multimedia, Software Engineering and Computing, vol 2. Springer, Berlin, Heidelberg, pp 471–476

De Mello LH, Sanderson AC (1990) AND/OR graph representation of assembly plans. IEEE Trans Robot 6(2):188–199

Dehghani M, Hubálovský Š, Trojovský P (2021) Northern goshawk optimisation: a new swarm-based algorithm for solving optimisation problems. IEEE Access 9:162059–162080

Fathollahi-Fard AM, Woodward L, Akhrif O (2021) Sustainable distributed permutation flow-shop scheduling model based on a triple bottom line concept. J Ind Inf Integr 24:100233

Feng Y, Gao Y, Tian G, Li Z, Hu H, Zheng H (2018) Flexible process planning and end-of-life decision-making for product recovery optimisation based on hybrid disassembly. IEEE Trans Autom Sci Eng 16(1):311–326

Feng Y, Zhang Z, Tian G, Fathollahi-Fard AM, Hao N, Li Z, Wang W, Tan J (2019) A novel hybrid fuzzy grey TOPSIS method: supplier evaluation of a collaborative manufacturing enterprise. Appl Sci 9(18):3770.4

Gholizadeh H, Fazlollahtabar H, Fathollahi-Fard AM, Dulebenets MA (2021) Preventive maintenance for the flexible flowshop scheduling under uncertainty: a waste-to-energy system. Environ Sci Pollut Res:1–20

Güngör A, Gupta SM (2001) Disassembly sequence plan generation using a branch-and-bound algorithm. Int J Prod Res 39(3):481–509

Guo X, Zhang Z, Qi L, Liu S, Tang Y, Zhao Z (2021) Stochastic hybrid discrete grey wolf optimizer for multi-objective disassembly sequencing and line balancing planning in disassembling multiple products. IEEE Trans Autom Sci Eng

Henrioud JM, Bourjault A (1991) LEGA: a computer-aided generator of assembly plans. In: Computer-aided Mechanical Assembly Planning. Springer, Boston, MA, pp 191–215

Laili Y, Wang Y, Fang Y, Pham DT (2021) Optimisation of Robotic Disassembly for Remanufacturing. Springer Cham, Switzerland, p 183 ISBN 978-3-030-81798-5.

Lambert AJD, Gupta SM (2008) Methods for optimum and near optimum disassembly sequencing. Int J Prod Res 46(11):2845–2865

Liu Y, Jiang Z, Yu J (2021) Human-machine collaboration-based disassembly sequence planning for power cell systems. Combined Machine Tools and Automated Machining Technology

Ma F, Zhang H, Gong Q, Hon KKB (2021) A novel energy efficiency grade evaluation approach for machining systems based on inherent energy efficiency. Int J Prod Res 59(19):6022–6033

McGovern SM, Gupta SM (2006) Ant colony optimisation for disassembly sequencing with multiple objectives. J Adv Manuf Technol 30(5):481–496

Moore KE, Güngör A, Gupta SM (2001) Petri net approach to disassembly process planning for products with complex AND/OR precedence relationships. Eur J Oper Res 135(2):428–449

Subramani AK, Dewhurst P (1991) Automatic generation of product disassembly sequences. CIRP Annals 40(1):115–118

Suga, T., Saneshige, K., & Fujimoto, J. (1996). Quantitative disassembly evaluation. In Proceedings of the 1996 IEEE International Symposium on Electronics and the Environment. ISEE-1996 (pp. 19-24). IEEE.

Tan WJ, Chin CMM, Garg A, Gao L (2021) A hybrid disassembly framework for disassembly of electric vehicle batteries. Int J Energy Res 45(5):8073–8082

Tian G, Chu J, Jin X, Zhang T, Wang H (2011a) Probability-based evaluation method and mathematical description of product detachability. Comput Integr Manuf Syst 17(6):1164–1170

Tian G, Chu J, Jin X, Zhang T, Wang H (2011b) Probability-based evaluation method and mathematical description of product detachability. Comput Integr Manuf Syst 17(6):1164–1170

Tian G, Liu Y, Ke H, Chu J (2012) Energy evaluation method and its optimisation models for process planning with stochastic characteristics: a case study in disassembly decision-making. Comput Ind Eng 63(3):553–563

Tian G, Zhou M, Li P (2017) Disassembly sequence planning considering fuzzy component quality and varying operational cost. IEEE Trans Autom Sci Eng 15(2):748–760

Tian G, Yuan G, Aleksandrov A, Zhang T, Li Z, Fathollahi-Fard AM, Ivanov M (2022a) Recycling of spent lithium-ion batteries: a comprehensive review for identification of main challenges and future research trends. Sustain Energy Technol Assess 53:102447

Tian G, Zhang C, Fathollahi-Fard AM, Li Z, Zhang C, Jiang Z (2022b) An enhanced social engineering optimiser for solving an energy-efficient disassembly line balancing problem based on bucket brigades and cloud theory. IEEE Trans Industr Inform

Wang W, Tian G, Chen M, Tao F, Zhang C, Abdulraham A, Li Z, Jiang Z (2020) Dual-objective program and improved artificial bee colony for the optimisation of energy-conscious milling parameters subject to multiple constraints. J Clean Prod 245:118714

Wang W, Tian G, Zhang T, Jabarullah NH, Li F, Fathollahi-Fard A (2021) Scheme selection of design for disassembly (DFD) based on sustainability: a novel hybrid of interval 2-tuple linguistic intuitionistic fuzzy numbers and regret theory. J Clean Prod 281:124724

Wolpert DH, Macready WG (1997) No free lunch theorems for optimisation. IEEE Trans Evol Comput 1(1):67–82

Wu T, Zhang Z, Yin T, Zhang Y (2022) Multi-objective optimisation for cell-level disassembly of waste power battery modules in human-machine hybrid mode. Waste Manag 144:513–526

Xiao J, Anwer N, Li W, Eynard B, Zheng C (2022) Dynamic Bayesian network-based disassembly sequencing optimisation for electric vehicle battery. CIRP J Manuf Sci Technol 38:824–835

Xie J, Li X, Gao L (2021) Disassembly sequence planning based on a modified grey wolf optimiser. J Adv Manuf Technol 116(11):3731–3750

Yu J, Zhang H, Jiang Z, Yan W, Wang Y, Zhou Q (2022) Disassembly task planning for end-of-life automotive traction batteries based on ontology and partial destructive rules. J Manuf Syst 62:347–366

Yuan G, Yang Y, Tian G, Zhuang Q (2020) Comprehensive evaluation of disassembly performance based on the ultimate cross-efficiency and extension-gray correlation degree. J Clean Prod 245:118800

Yuan G, Yang Y, Tian G, Fathollahi-Fard AM (2022) Capacitated multi-objective disassembly scheduling with fuzzy processing time via a fruit fly optimisation algorithm. Environ Sci Pollut Res:1–18

Zeng Y, Zhang Z, Yin T, Zheng H (2022) Robotic disassembly line balancing and sequencing problem considering energy-saving and high-profit for waste household appliances. J Clean Prod 135209

Zhang C, Fathollahi-Fard AM, Li J, Tian G, Zhang T (2021) Disassembly sequence planning for intelligent manufacturing using social engineering optimiser. Symmetry 13(4):663

Zhu JF (2020) Research on disassembly sequence planning and target inspection of electromechanical products for remanufacturing (Master's thesis). Shandong University